

RISK-02-TECHNICAL-RISK-ANALYSIS

RISK-02: DEEP DIVE — TECHNICAL RISK ANALYSIS

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I. EXECUTIVE SUMMARY

Every advanced materials company faces the gap between laboratory validation and field deployment. This document provides a deep-dive analysis of the 9 most significant technical risks facing Carrington Storm Motors' materials and products. For each, we present current evidence, remaining uncertainty, testing plan to de-risk, and fallback options.

Under the Accountant Heuristic: we do not minimize these risks. They are real. We present them fully because investors who understand the risks can evaluate whether the mitigation is adequate. Investors who are shielded from the risks cannot.

II. TECHNICAL RISK DEEP DIVES

Risk 1: MXene Stability and Longevity Beyond 5,000 Hours

Description: MXenes are a relatively young materials class (discovered 2011). The longest-duration published stability study for MXene-polymer composites is approximately 3,000 hours under accelerated aging conditions. CSM's internal data extends to 5,000 hours. Utilities require 40-year service life. The gap between 5,000 hours (approximately 7 months) and 40 years (350,000 hours) is bridged by Arrhenius extrapolation, not measurement.

Current Evidence: - Accelerated aging at 85°C/85% relative humidity for 5,000 hours: EMI SE degradation of 1.2-3.8% depending on polymer matrix. Polypropylene matrix shows best stability (1.2% degradation); PVDF shows worst (3.8%). - Arrhenius model using activation energy derived from degradation rates at 65°C, 85°C, and 100°C projects 40-year SE retention of 88-94% (for PP-matrix composite). Model $R^2 = 0.91$. - XRD analysis of aged specimens shows no detectable oxidation of MXene nanosheets (no TiO₂ peaks), suggesting degradation mechanism is polymer matrix chain scission rather than MXene oxidation, which is favorable for long-term stability. - Data in: Materials Science Vol. 2, Appendices E-H.

Remaining Uncertainty: - Arrhenius models assume a single dominant degradation mechanism with constant activation energy. If additional degradation mechanisms activate at longer timescales (e.g., slow interfacial debonding between MXene and polymer, stress-corrosion cracking under combined thermal-mechanical loading), the Arrhenius projection may be optimistic. - The 5,000-hour data set is from laboratory-prepared specimens with high QC. Field-manufactured composites may have initial defect populations (voids, agglomerations) that accelerate degradation. - Synergistic effects of combined stressors (UV + moisture + thermal cycling + mechanical load + GIC current) cannot be replicated in single-stressor accelerated aging. Multi-stressor aging testing is planned but has not been conducted.

Testing Plan to De-Risk: - Continue accelerated aging to 10,000 hours (target: Q4 2027). If Arrhenius projection holds at 10,000 hours, confidence in the model increases significantly. - Initiate multi-stressor aging (combined UV + moisture + thermal cycling) in Q1 2027. - Field-deploy MXene composite witness coupons at all 5 pilot utility sites. Coupons retrieved at 6, 12, 24, and 36 months and characterized for SE, mechanical properties, and chemical composition. - Independent materials characterization at NIST (Boulder) to validate internal measurements. - Initiate modeling of MXene-polymer interfacial degradation using molecular dynamics simulation (DOE INCITE or NSF XSEDE computational allocation).

Fallback Options: 1. **Enhanced Stabilizer Package:** If degradation rate is higher than projected, reformulate with additional anti-oxidants and UV stabilizers. This is a formulation adjustment, not a fundamental redesign. Estimated impact: 5-10% increase in material cost, 5-8% reduction in SE (stabilizers are non-conductive and

dilute MXene loading). 2. **MXene Passivation:** Surface-functionalize MXene nanosheets to improve oxidation resistance. Our A-4 invention disclosure covers this approach. Estimated timeline: 12-18 months to develop and validate. 3. **Gen 2 MXene (Double Transition Metal MXenes):** Mo-based or Nb-based MXenes may have inherently higher oxidation resistance than Ti-based MXenes. These are at TRL 2-3 (early academic research). This is a long-term fallback, not a near-term fix. 4. **Warranty Adjustment:** If long-term stability cannot be projected to 40 years, offer 15-20 year warranty with scheduled MXene layer replacement as a maintenance service. This is commercially viable but less attractive to utilities accustomed to 40-year asset lifetimes. 5. **Accept Risk:** Even if MXene degrades over 40 years, the product provides protection during the highest-risk period (the next 20 years, when a Carrington Event is statistically most likely to occur before the grid is otherwise hardened). This argument is correct but may not satisfy utility procurement requirements.

Risk 2: CNT-Polymer Manufacturing Consistency at Scale

Description: Achieving consistent dispersion of carbon nanotubes in a polymer matrix at industrial scale is one of the most persistent challenges in nanocomposite manufacturing. CNTs tend to agglomerate due to van der Waals forces, creating conductive pathways that are unevenly distributed, which produces inconsistent EMI shielding effectiveness.

Current Evidence: - Laboratory-scale (500g batch) extrusion runs (n=22) achieved SE variation of ± 2.1 dB at 2.5 wt% CNT loading. - TEM imaging confirms good dispersion (no agglomerates $> 5 \mu\text{m}$ observed in QC specimens). - However, these batches were produced on a benchtop extruder under carefully controlled conditions by Dr. Chen personally. They are not representative of production-scale consistency.

Remaining Uncertainty: - Scaling from 500g lab batches to 50+ kg production batches introduces new variables: screw wear affecting shear profile, feedstock moisture content variation, lot-to-lot CNT quality variation, operator variability. - The percolation threshold for this composite is approximately 0.8 wt% CNT. We operate at 2.5 wt% — well above percolation — which provides a buffer against dispersion inconsistency. If dispersion is poor (agglomerated CNTs), the effective percolation network is less dense, and SE drops. - CNT suppliers: 4 qualified, but supplier qualification at kilogram scale has not been validated at ton scale. A supplier change could affect dispersion quality.

Testing Plan to De-Risk: - Pilot plant extrusion runs at incrementally larger batch sizes: 5 kg, 25 kg, 50 kg, 100+ kg. SE measured on 10 specimens per batch. Statistical process control limits established based on initial 20-batch dataset. - In-line conductivity measurement (4-point probe immediately post-extrusion) to detect dispersion issues in real time, before downstream processing. - CNT supplier qualification: 50 kg qualification batches from each of 4 suppliers, with full SE, mechanical, and dispersion characterization. - Design of experiments (DOE) to identify critical process parameters affecting dispersion: screw speed, barrel temperature profile, feed rate, CNT pre-dispersion method.

Fallback Options: 1. **Pre-Dispersed CNT Masterbatch:** Purchase CNTs pre-dispersed in a carrier polymer from a specialty supplier (several exist: Hyperion Catalysis, Nanocyl, OCSiAl). This transfers dispersion risk to the supplier but increases material cost by 40-80% and introduces supply chain dependency. 2. **Reduce CNT Loading:** If dispersion drives inconsistency, reduce CNT loading to 1.5 wt% (still above percolation) and accept 10-15% lower SE. This requires thicker composite for equivalent SE, increasing material usage and weight. 3. **Switch to Graphene Nanoplatelets:** GNPs have less tendency to agglomerate than CNTs and provide comparable electrical conductivity above 5 wt% loading. Trade-off: GNPs are less effective at low loading (percolation threshold $\sim 3-4$ wt%) and may be more expensive. But GNP dispersion is more forgiving at industrial scale. 4. **Accept Higher QC Reject Rate:** Implement 100% QC on production batches, rejecting batches that fall below SE specification. This increases cost (reject rate $\times$ material cost) but ensures product quality. Acceptable if reject rate is $< 10\%$.

Risk 3: Ceramic Matrix Composite (CMC) Brittle Failure

Description: SiC-fiber-reinforced Al_2O_3 CMC has high compressive strength but relatively low fracture toughness ($4-6 \text{ MPa}\cdot\text{m}^{1/2}$ compared to $50+ \text{ MPa}\cdot\text{m}^{1/2}$ for steel). While the transformer housing is loaded primarily in compression (where CMC performs excellently), any tensile or impact loading could cause catastrophic brittle fracture without warning.

Current Evidence: - Compression testing: 450 MPa (± 22 MPa, n=31). Excellent. - Flexural testing (4-point bend): 180 MPa (± 15 MPa, n=24). Adequate. - Fracture toughness (SENB): $5.2 \text{ MPa}\cdot\text{m}^{1/2}$ (± 0.8 , n=18). Low by structural standards. - Impact testing (Charpy): 3.8 kJ/m^2 . Low. A steel housing would absorb 10-20x more impact energy. - Acoustic emission monitoring during compressive loading shows no significant pre-failure acoustic

activity — the material fails suddenly (“graceful failure” is not a characteristic).

Remaining Uncertainty: - Long-term subcritical crack growth: does slow crack propagation occur under sustained load (transformer weight + thermal cycling), eventually leading to failure below the design load? We have stress-rupture data only to 1,000 hours. Utilities want 40-year creep data. - Damage tolerance: if a CMC housing is impacted during transport or installation (e.g., dropped tool, vehicle collision), how does the resulting flaw affect long-term strength? We need damage tolerance characterization (post-impact residual strength testing). - Weibull modulus: the statistical distribution of strength in brittle materials. A low Weibull modulus means wide variability, requiring higher design safety factors. Our Weibull modulus data is preliminary (n=31).

Testing Plan to De-Risk: - Expand the mechanical test matrix: increase specimen count to $n \geq 50$ for baseline properties, improving Weibull modulus confidence. - Damage tolerance testing: instrumented impact at 5, 10, 25, 50 J; post-impact residual compressive strength measurement. - Stress-rupture (creep) testing: sustained load at 60%, 70%, 80% of UTS at elevated temperature for accelerated testing. Target: 5,000-hour data within 18 months. - Non-destructive evaluation development: optimize ultrasonic C-scan parameters to detect sub-millimeter flaws. Validate NDE sensitivity with intentional flaw specimens. - Full-scale prototype testing: fabricate a ShieldCore housing at 1:3 scale and test to failure. Instrument with strain gauges, acoustic emission, and digital image correlation.

Fallback Options: 1. **Increased Safety Factor:** If Weibull modulus is lower than expected (indicating wide strength variability), increase design safety factor from 3x to 5x, reducing allowable stress but ensuring statistical reliability. This increases material usage and housing weight by 40-60% but is technically straightforward. 2. **Fiber Architecture Optimization:** If fracture toughness is the primary concern, increase fiber volume fraction from current 35% to 45-50% and optimize fiber architecture (3D woven preform instead of 2D layup) for improved toughness. This increases manufacturing cost by 25-35%. 3. **Hybrid CMC-Steel Housing:** If damage tolerance is critical, design a steel exoskeleton with CMC interior housing — steel provides impact resistance and damage tolerance, CMC provides electrical insulation. This is heavier and more expensive but technologically conservative. Best fallback for utility acceptance. 4. **External MXene Composite Wrap:** Apply MXene-polymer composite overwrap to CMC housing for crack arrest capability. The MXene composite is tougher (higher elongation) and can bridge cracks in the CMC. This is a lightweight, low-cost toughness enhancement.

Risk 4: Geopolymer Concrete Curing Variability in Field Conditions

Description: Laboratory geopolymer specimens achieve 65 MPa at 28 days under controlled curing (23°C, 95% RH). Field-cast geopolymer in variable weather may not achieve design strength, particularly in cold conditions (Montana winters, where substation work is often done in off-peak seasons).

Current Evidence: - Laboratory curing at 23°C: 65 MPa at 28 days (n=28). - Laboratory curing at 5°C: 32 MPa at 28 days — 50% strength reduction. - Laboratory curing at 35°C: 82 MPa at 28 days — accelerated curing. - Fly ash composition variability: 3 fly ash sources tested; strength variation ± 8 MPa at constant mix design. - Freeze-thaw resistance: 300+ cycles with <5% mass loss (ASTM C666). Excellent.

Remaining Uncertainty: - Combined cold + variable fly ash: worst-case combination of cold weather and high-calcium fly ash (accelerates setting, potentially causing cold joints in large pours) has not been tested. - Field QC methods: cylinder testing is the standard but cylinders cured on-site may not represent in-place concrete due to different thermal history (mass effect in large pours). - Long-term electrical resistivity: does resistivity degrade over time as the geopolymer matrix interacts with soil moisture and groundwater? We have 2-year resistivity data (stable); 40-year extrapolation is based on mechanism understanding (no sulfate or chloride ingress observed), not measurement.

Testing Plan to De-Risk: - Cold-weather admixture development: formulate an accelerator admixture for geopolymer concrete, analogous to calcium chloride for Portland cement but compatible with geopolymer chemistry. This is a significant R&D target. - Field trial pours: cast test foundations at Montana facility in winter conditions (December-February). Instrument with thermocouples to measure in-place temperature profile. Test cores at 7, 28, and 90 days. - Expanded fly ash qualification: test 6 additional fly ash sources for composition-property correlation. Develop a fly ash acceptance protocol (XRF composition limits, particle size distribution, LOI limits). - Long-term resistivity monitoring: install resistivity probes in pilot substation foundations at 5 utility sites. Measure monthly for 3 years. - Maturity method calibration: calibrate the maturity method (ASTM C1074) for strength estimation of in-place geopolymer concrete, enabling real-time strength assessment without waiting for cylinder breaks.

Fallback Options: 1. **Precast Only:** If field curing is unreliable, manufacture all geopolymer elements as

precast units in a controlled factory environment. Accept higher transportation cost and crane requirements on-site. 2. **Heated Formwork:** For cold-weather field pours, use heated formwork (electric heating blankets or embedded heating elements) to maintain 20°C minimum during the first 72 hours of curing. Adds cost but is proven technology. 3. **Portland Cement Backup:** If geopolymer cannot meet schedule or strength requirements in the field, fall back to high-resistivity Portland cement concrete (achievable with silica fume addition, achieving $10^5 \Omega\cdot\text{cm}$ — still 3 orders of magnitude lower than geopolymer but better than standard concrete). This is not ideal but is always available. 4. **Hybrid Foundation:** Composite foundation with geopolymer base course (precast) and Portland cement overlay. The geopolymer provides the dielectric layer; the Portland cement provides constructability. Less elegant but pragmatically robust.

Risk 5 through Risk 9 (Summarized)

Risk 5 — YInMn Blue Batch-to-Batch Color Consistency: - Evidence: 8 pigment batches showed threshold temperature variation of $\pm 3.2^\circ\text{C}$ (target: $\pm 2^\circ\text{C}$). - Mitigation: automated precursor weighing, in-line spectrophotometry, QC rejection of out-of-spec batches. - Fallback: wider specification ($\pm 5^\circ\text{C}$) acceptable for “caution” indication; still useful even if imprecise.

Risk 6 — Protonic Transceiver Range (TRL 3 → TRL 6 Gap): - Evidence: 12 km demonstrated. 120 km target is 10x improvement. - Mitigation: staged development (30 → 60 → 120 km), relay drone mesh as architecture option. - Fallback: accept relay-drone-based architecture as primary design rather than point-to-point 120 km. Acceptable trade-off.

Risk 7 — Amphibious Walker Mechanical Reliability at 8-Ton Scale: - Evidence: No prior art at this scale. Legged robotics proven at 1-100 kg scale (Boston Dynamics, ANYbotics). 8 tons is unprecedented. - Mitigation: conservative design (2x safety factor), hydraulic over electric actuation, extensive test program. - Fallback: conventional amphibious wheeled vehicle (proven technology) with MXene-shielded electronics. Surrenders the ability to traverse debris fields but can still reach most coastal substations via water.

Risk 8 — MXene Synthesis Yield at Scale: - Evidence: Lab yield ~40%. Continuous-flow target >85%. First continuous-flow prototype achieved 52%. - Mitigation: highest-priority R&D program; multiple parallel development paths. - Fallback: accept lower yield (higher MXene cost); outsource MXene production; pursue electrochemical exfoliation.

Risk 9 — Combined Stressor Degradation Not Characterized: - Evidence: Only single-stressor accelerated aging data available. - Mitigation: multi-stressor aging program (Q1 2027 initiation); field witness coupons at pilot sites. - Fallback: conservative design margin (2x material thickness above SE requirement); regular field inspection and replacement program.

III. TECHNICAL RISK TREND ANALYSIS

The technical risk profile is front-loaded: the highest uncertainties are in the next 24 months (pilot plant scale-up, certification testing, initial field deployments). As these uncertainties are resolved — either validated or mitigated — the technical risk profile declines significantly. By Year 3, assuming pilot milestones are met, the technical risk register should have no risks scored above 9.

The specific inflection points: - MXene continuous-flow yield $\geq 75\%$: downgrades T-08 from 12 to 6 - CMC damage tolerance characterization complete and acceptable: downgrades T-03 from 12 to 6 - Protonic transceiver achieves 60 km in field test: downgrades T-06 from 12 to 8 - Amphibious walker completes 500-hour test stand endurance: downgrades T-07 from 16 to 8 - 10,000-hour MXene aging data validates Arrhenius projection: downgrades T-01 from 8 to 4

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